

MAHESHCHANDRA HEGDE

Fractional CTO | Technical Architect | Creative Technologist

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EXECUTIVE PROFILE & SUMMARY

Versatile Technology Leader, Principal Architect, and Product Strategist with **18+ years** of track record architecting high-performance digital products, distributed cloud platforms, and immersive spatial 3D systems. Expert in **React, Angular, TypeScript, Node.js, Web Components (LitElement), Cloud architectures (GCP/Firebase, AWS)**, and resilient offline-first PWAs. Dual-disciplined background blending Human Interface Design (**M.S. in Computing, UK**) with Electronics & Communication Engineering (**B.E.**), uniquely bridging high-level executive business strategy with deep, hands-on architectural rigor.

Currently serving as **Fractional CTO at eBodhya Technologies** (leading AI-assisted Academic Intelligence & school management systems) and **Founder & CTO at Technoyana Digital Transformation Services**, incubating deep-tech product studios including **SrushtiLabs** (generative 3D asset platforms, Voxelforge AI, Ayam3d) and **Twitan** (high-reliability sports tournament management OS, Shutlify & Twicket). Proven background executing mission-critical enterprise modernizations (4-year zero-downtime migration of Cisco Stadium Vision Director), clinical ICU software (Philips Healthcare ICCA), and cross-framework UI design systems. Available for fractional leadership, architecture governance, and technical consulting.

CORE COMPETENCIES & TECHNICAL MASTERY

Executive & Technical Leadership Fractional CTO, Technical Architecture, System Modernization, Product Strategy, Agile/Scrum, Mentorship, Engineering Roadmaps	Frontend & Web Systems React, TypeScript, Angular, Web Components (LitElement), Next.js, Vite, Redux, Modern JavaScript (ES6+), HTML5/CSS3, WCAG 2.1 AA
AI, Spatial Computing & 3D Tech Generative 3D Asset Creation, Mesh Synthesis, Automated Retopology, Tileable PBR Textures & Upscaling, Three.js, WebGL, 3ds Max, Blender	Cloud, Backend & Offline Architecture Node.js, Express, Firebase / Google Cloud Platform (GCP), AWS, REST APIs, Micro-frontends, PWA (Offline-first / IndexedDB)
Enterprise Security & Reliability Strict Healthcare Security (MFA, RBAC, XSS Prevention), HIPAA/PHI Data Protection, Zero Memory-Leak Continuous Clinical Uptime	Design Systems & UX Strategy Enterprise UI Design Systems, Design-to-Code Automation, Interactive Prototyping, Figma, Adobe XD, Photoshop, Human-Interface Design

PROFESSIONAL EXPERIENCE

- Fractional Chief Technology Officer (CTO) — eBodhya Technologies Private Limited**
Bengaluru, India · Academic Intelligence Operating System & EdTech SaaS Platform

2025 – Present

 - Drive technical leadership and architectural vision for an interconnected Academic Intelligence OS designed to streamline school operations, curriculum workflows, and assessment processes.
 - Architect curriculum-aware AI workflows and evaluation engines linking question banks, automated grading, and real-time student analytics into a unified source of truth.
 - Establish modular full-stack architecture, scalability guidelines, and automated deployment pipelines, empowering educators and accelerating product-market rollout.
- Founder & Chief Technology Officer (CTO) — Technoyana Digital Transformation Services Pvt. Ltd.**
Bengaluru, India · Enterprise Product Engineering, Spatial AI & High-Reliability SaaS Studio

May 2021 – Present

 - Direct end-to-end technology vision, cloud architecture, and product engineering across Technoyana's incubator portfolio and enterprise consulting initiatives.
 - SrushtiLabs (Spatial AI & 3D Tech Division) | srushtilabs.com:**
 - Architected **Voxelforge AI** (srushtilabs.com/voxelforge/), a generative 3D asset platform producing modular, low-poly 3D models and asset bundles optimized for real-time engines including Unreal Engine, Unity, and WebGL.
 - Engineered **Ayam3d** (ayam3d.in/), a parametric AI 3D mesh synthesis engine transforming natural language prompts into production 3D assets with automated retopology and tileable PBR texture pipelines; secured early investor validation.
 - Twitan (Sports SaaS Studio) | twitan.com:**
 - Architected **Shutlify (Badminton OS)**, an end-to-end tournament management platform featuring automated knockout/round-robin fixture engines, live court arbitration, multi-court scheduling, and offline-first PWA resilience for unreliable venue networks (deployed at tournaments like HBL Sirsi).
 - Developed **Twicket**, a high-fidelity real-time scoring and match statistics engine for cricket tournaments.
 - Specialized Multi-Disciplinary Engineering:** Designed customized CNC machine controller interfaces and automated CAD/CAM toolpath generation pipelines; delivered architectural 3D simulations and consumer IoT parking payment applications.

Senior Technical Lead (React / Node.js) — Cyient Limited / Philips Healthcare

Nov 2023 – Mar 2025

Bengaluru, India · Philips IntelliSpace Critical Care & Anesthesia (ICCA) · Mission-Critical ICU Software

- Led web-tier engineering and frontend architecture for Philips ICCA clinical software deployed in hospital ICUs and anesthesia departments globally.
- Ensured 24/7 continuous clinical reliability with zero memory leaks and sub-second UI responsiveness under dense, real-time streaming patient telemetry.
- Strictly enforced enterprise healthcare security policies including Multi-Factor Authentication (MFA), Cross-Site Scripting (XSS) prevention, granular Role-Based Access Control (RBAC), and strict patient data privacy compliance.
- Governed code quality, architectural standards, automated regression testing, and modular micro-frontends across distributed agile squads.

Development Lead / Senior Analyst — Ness Digital Engineering

Oct 2019 – Apr 2020

Bengaluru, India · Entertainment Management Application (Angular / AWS / Node.js)

- Architected the complete single-page application structure using Angular, TypeScript, Node.js, and AWS cloud infrastructure.
- Enforced strict accessibility standards (WCAG 2.1 AA), frontend security best practices, and code quality linters across the engineering squad.
- Took technical ownership of the CI/CD deployment pipelines, successfully delivering the application to client production on schedule.

User Interface Architect — Moonraft Innovation Labs (Unit of UST Global)

May 2019 – Jul 2019

Bengaluru, India · Enterprise UI Design System & Mobile Engineering

- Architected and engineered a centralized **Cross-Framework UI Design System** using **LitElement (Web Components), React, and Angular**, packaged and distributed via npm.
- Completely eliminated component duplication across multiple engineering squads, ensuring seamless brand coherence and automated upgradability.
- Led the mobile application development squad building a high-performance cross-platform guest booking app for **TLC Group of Hotels** using Ionic and Angular.

Associate Project Manager & Senior Systems Analyst — UST Global / Cisco Systems

Feb 2015 – Feb 2019

Bengaluru, India & Global Deployment · Cisco Stadium Vision Director (4+ Years Dedicated Tenure)

- Led the phased, zero-downtime legacy modernization of **Cisco Stadium Vision Director** from legacy Adobe Flash/Flex to modern **Angular and React** modules across premier international sporting stadiums.
- Architected hybrid micro-frontend bridges allowing legacy and modern components to coexist, preventing any disruption to live stadium operations.
- Managed cross-functional agile teams delivering single-page operational dashboards that substantially improved venue operational efficiency.
- Awarded three consecutive **UST Global Certificates of Excellence (2015, 2016, 2018)** for client-first commitment and outstanding engineering delivery.

UI Architect & Tech Lead — ThoughtFocus Technologies

Oct 2011 – Mar 2014

Bengaluru, India · Aftermarket Parts 3D Explorer & Enterprise Solutions

- Architected an enterprise Aftermarket Parts Explorer catalog supporting millions of industrial and automotive parts with high-speed search and filtering.
- Engineered an interactive 3D inspection viewer enabling buyers to explore, rotate, and measure 3D models of components prior to purchasing.
- Established rapid design-to-code prototyping pipelines; bootstrapped enterprise search systems and logistics management applications for a large milk/dairy corporation.

Independent Consultant (UX Designer & Full-Stack Developer) — Cisco Systems & Vodafone

Apr 2008 – Feb 2011

Bengaluru, India · Building Management Systems (BMS) Unified Dashboard

- Designed and developed the **Cisco Campus Unified Facility Dashboard**, integrating telemetry and control for AHUs, VRVs, industrial chillers, and campus environmental sensors.
- Awarded formal **Cisco Certificate of Appreciation (2010)** for outstanding operational visualization and intelligent facility telemetry. Delivered real-time data visualization dashboards for **Vodafone**.

Earlier Career & Foundational Experience

2004 – 2015

- **InnoBrik Software Technologies** — Co-Founder & Technical Leader (Feb 2011 – Jan 2015): Spearheaded product engineering, tech stack evaluation, and agile delivery across startup ventures and enterprise client engagements.
- **CAE Simulation Technologies** — Visual Database Developer (Sep 2007 – Apr 2008): Engineered 3D synthetic visual databases, topographical terrain models, and runway textures for real-time military and civil flight simulators under locked 60 FPS budgets.
- **Cognizant Technology Solutions** — Programmer Analyst Trainee (Apr 2004 – Dec 2004): Maintained mission-critical mainframe applications, COBOL routines, and DB2 queries for the DUNSLINK commercial credit platform.

EDUCATION

Master of Science (M.S.) in Computing

Robert Gordon University, Aberdeen, Scotland, UK (2005 – 2006)
Specialization: Human Interface Design & Development

Bachelor of Engineering (B.E.) in ECE

Bapuji Institute of Engineering & Technology (BIET), Davanagere, India (1999 – 2003)
Electronics & Communication Engineering

KEY HONORS & AWARDS

Cisco Certificate of Appreciation (2010)

Awarded for excellence in UX design and engineering of the Cisco Campus Unified BMS Facility Dashboard.

UST Global Certificates of Excellence (2015, 2016, 2018)

Triple recognition for technical leadership, innovation, and client delivery on Cisco Stadium Vision.